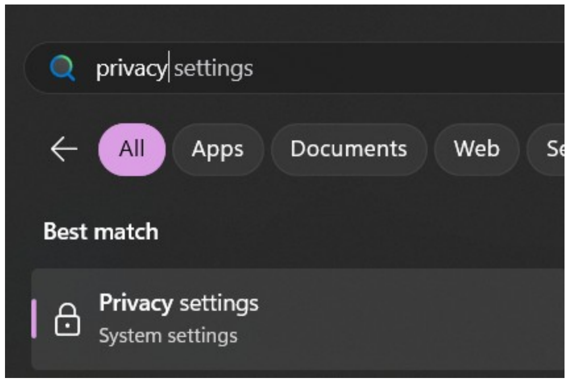
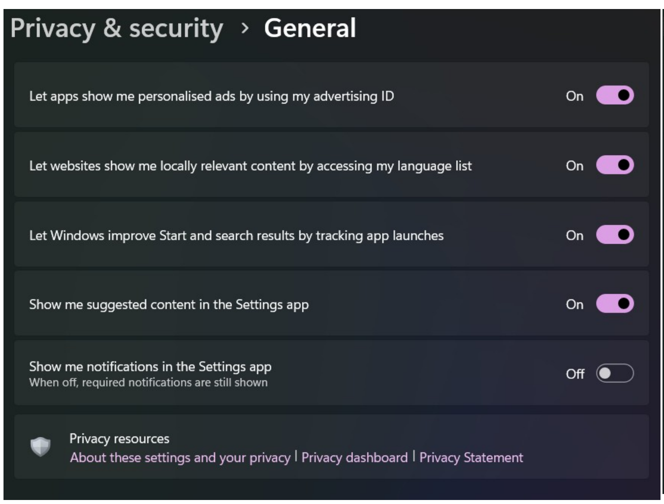
Lab 2: Advancing privacy past the default configurations

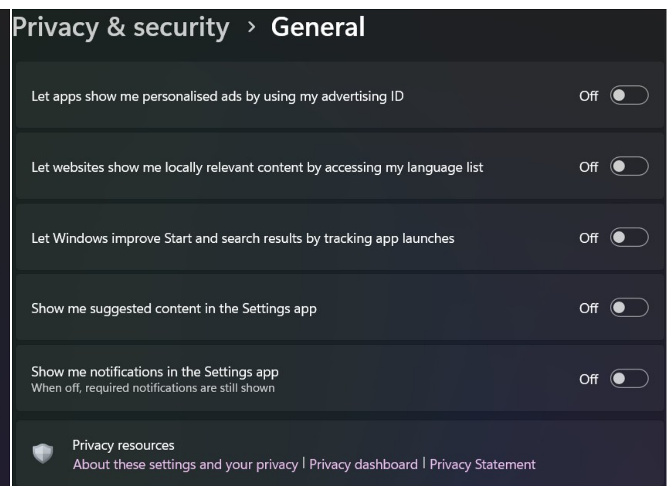
General Settings and Privacy

Step One - For the best privacy, it's recommended to press the Windows button on either the keyboard or on the taskbar to search for "**Privacy Settings**" to disable automatic tracking functions.



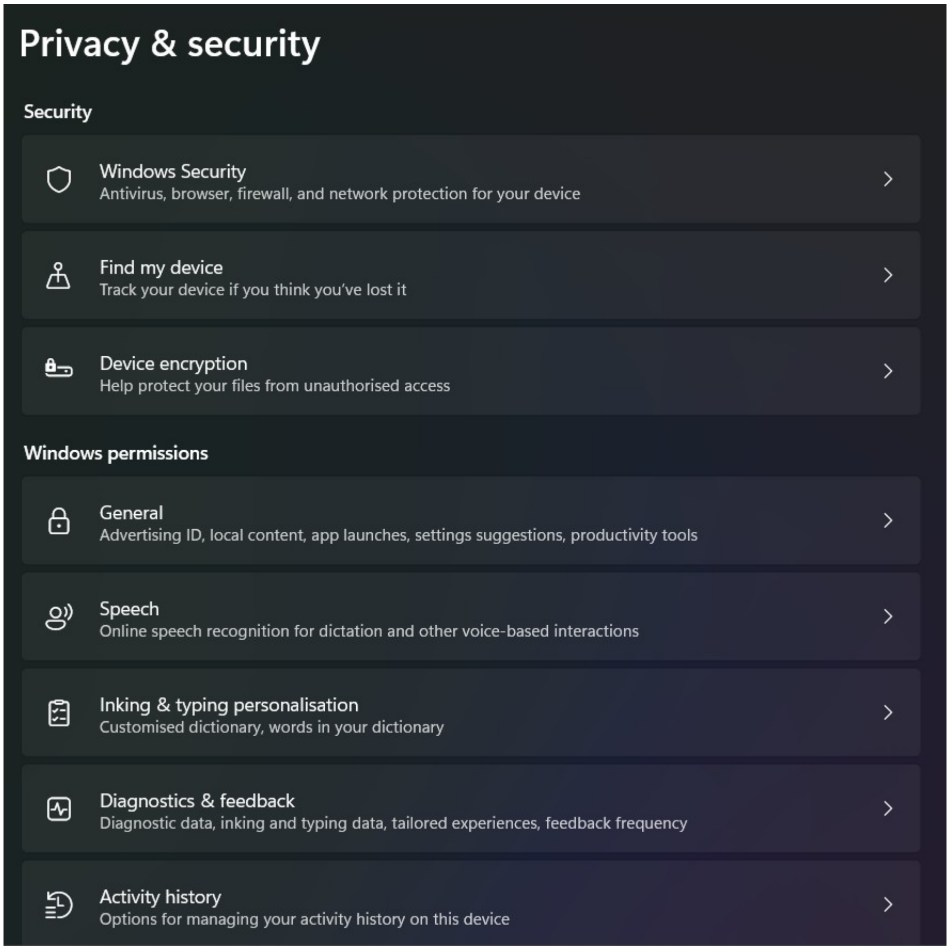
Step Two – You will see that each of these settings are automatically toggled on. You will see far less tracking from Microsoft once you click on each option to turn them off.





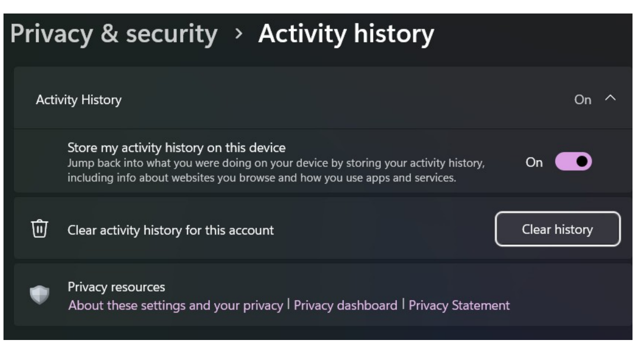
b.

Step Four – Click on the header that says “Privacy & Security”. This will show the main privacy setting menu. This section will be returned to frequently to ensure that each category is displaying the most privacy intensive settings.

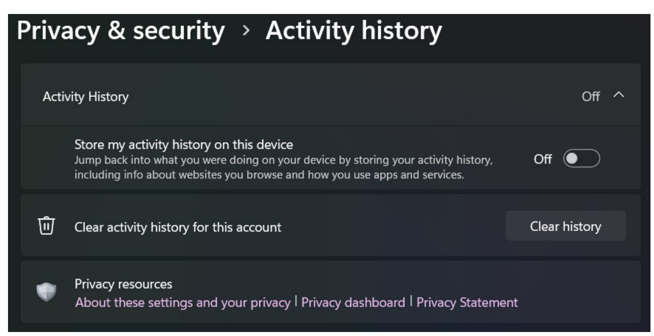


Step Five – Click on “**Activity history**” found near the bottom of the screen. Turn off the activity tracking by toggling the switch next to “**Store my activity history on this device**”

a.

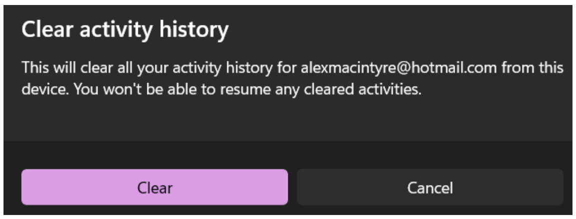


b.

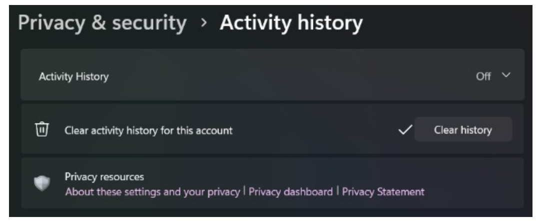


Step Six – Delete the activity history to ensure that no tracked activity was logged onto the computer. Press “**Clear history**”, which will prompt the user to confirm the action. Once the action is successfully completed, the “Clear history” button will have a checkmark beside it.

a.

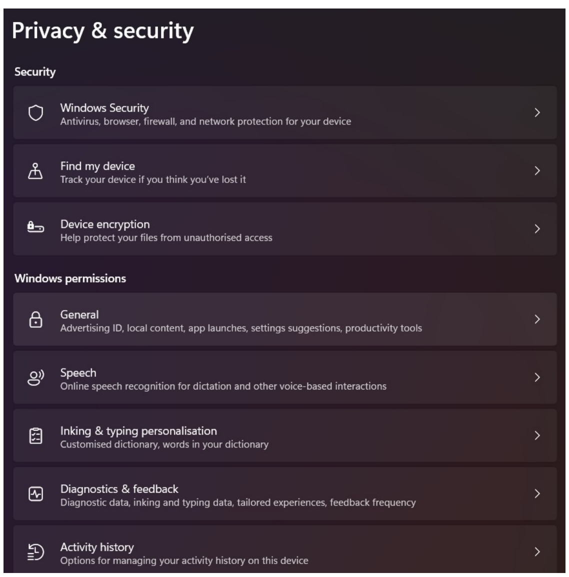


b.

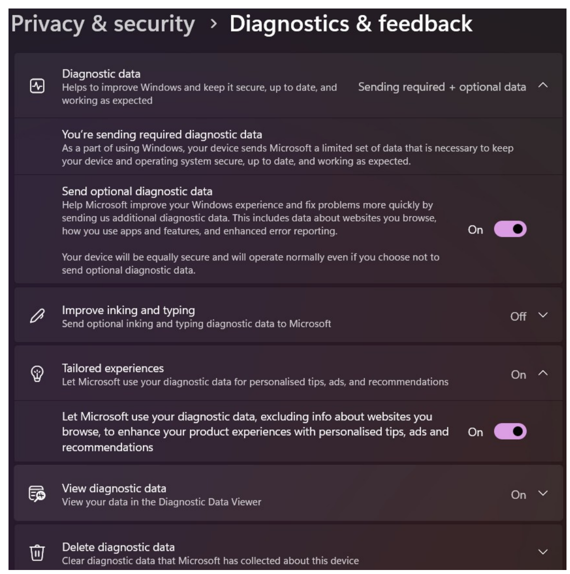


Step Seven – Return to the main Privacy & Security menu by clicking on the header, and go to the “**Diagnostics & Feedback**” section.

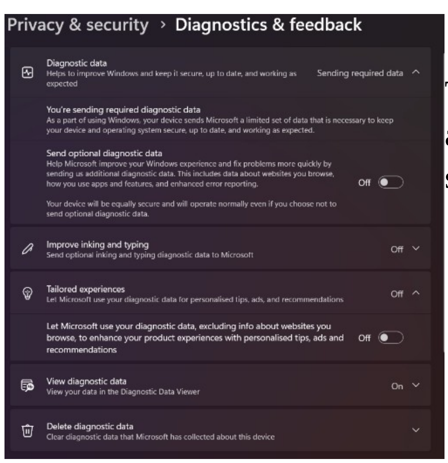
a.



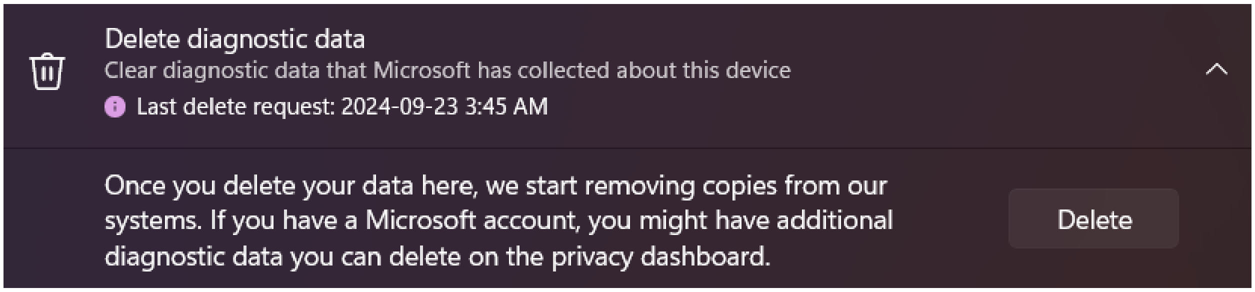
b.



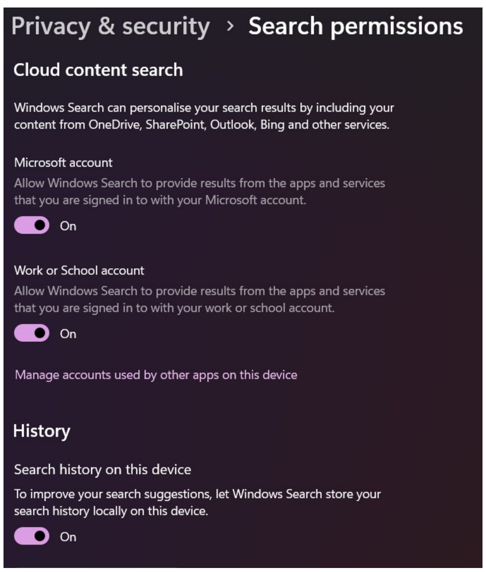
Step Eight – Toggle off the “**Send optional diagnostic data**" switch, alongside the switch in the “**Tailored experience**” section.

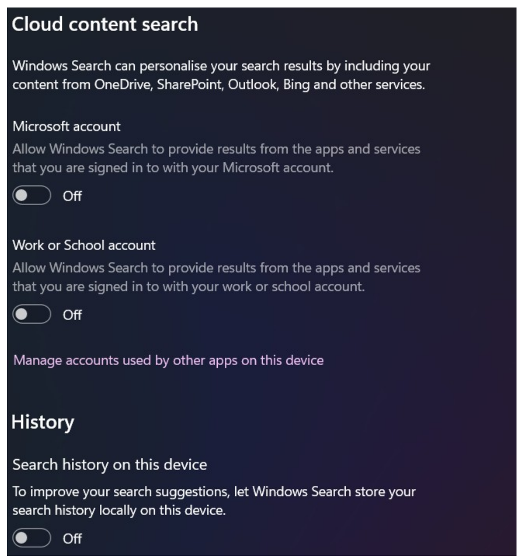


Step Nine – Near the bottom, click on "Delete diagnostic data", which will drop down and give you the option to delete the data. Once you click the "Delete" button, it will begin a request to remove the data from Microsoft's servers. Like the activity log, it will show a checkmark next to the delete button upon successful completion.

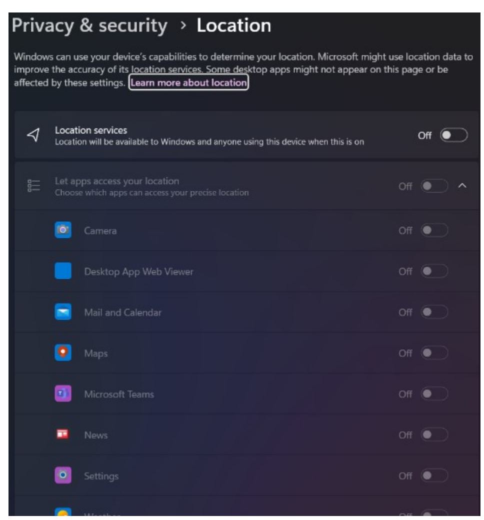


Step Ten – Return to the main Privacy & Security Menu from the previous steps using the header. Then, examine the Search Permissions section. Toggle off the switches for these settings.

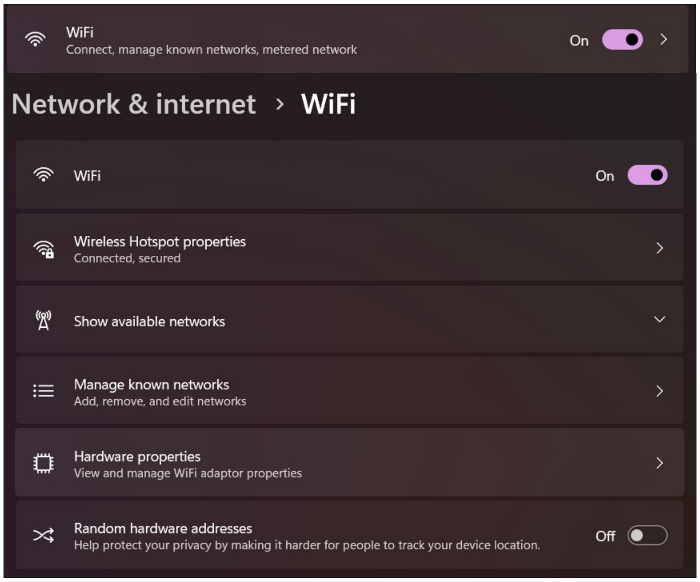


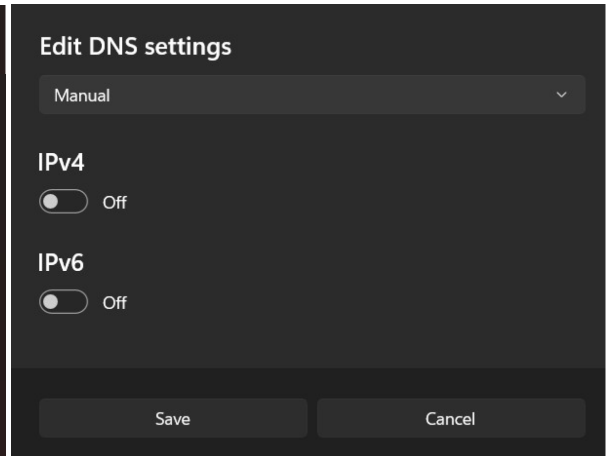


Step Eleven – Return to the main privacy settings menu and click on Location. Ensure that each option on this page is toggled off.



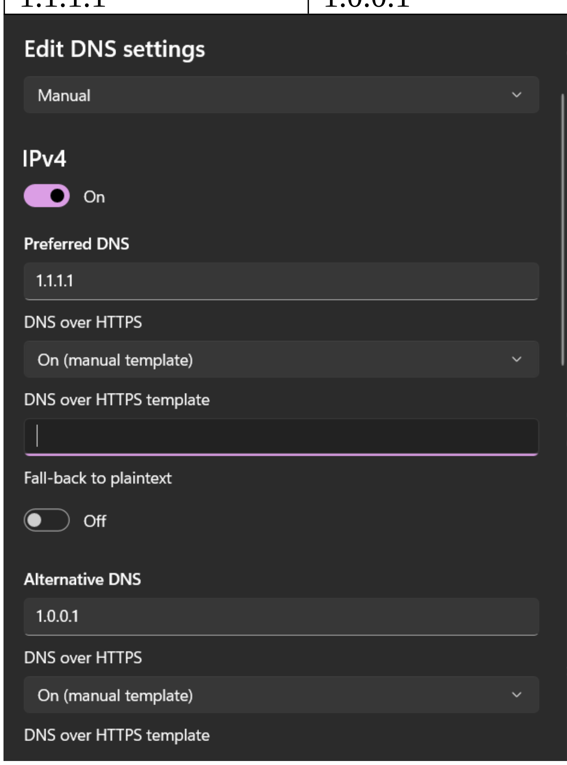
Step Eighteen – Return to “**Network & Internet**” and click on “**Wi-Fi**”. Once you are in the Wi-Fi menu, click on “**Hardware Properties**” Here, click “**Edit**” on the “**DNS Server Assignment**” button. This will allow you to change the DNS settings from Automatic (DHCP) to “**Manual**”.

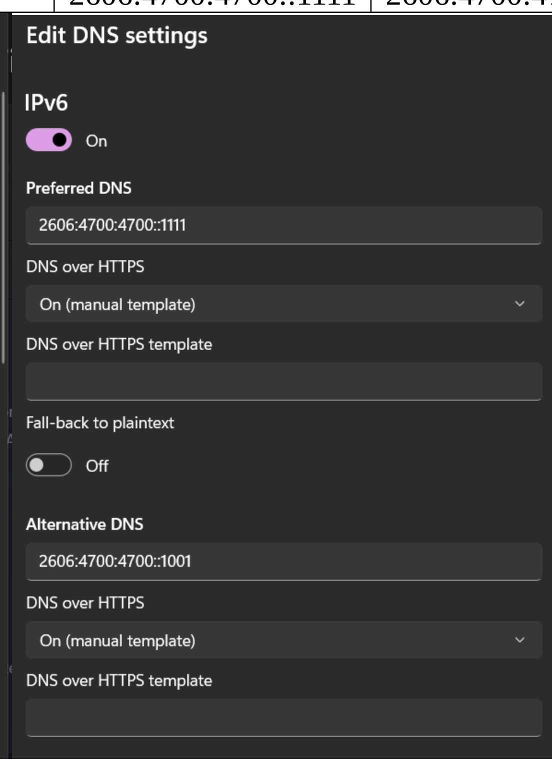




Step Nineteen – Now that the DNS setting is set to Manual, the IPV4 and IPV6 addresses can be toggled and changed using the Cloudflare DNS. This will provide a more private connection.

IPV4 Primary	IPV4 Secondary	IPV6 Primary	IPV6 Secondary
1.1.1.1	1.0.0.1	2606:4700:4700::1111	2606:4700:4700::1001





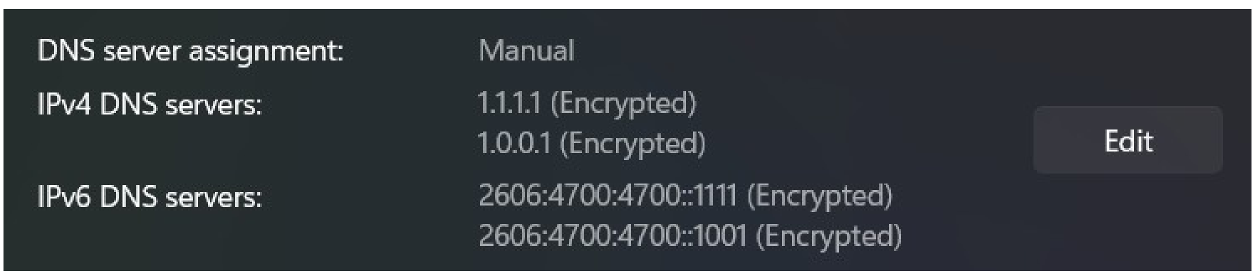
Step Twenty – Now that the addresses have been entered, DNS to HTTPS can be turned on through the dropdown menu for each of the four addresses as shown here:

Step Twenty One – In the field "DNS over HTTPS template" enter this link into all four fields:

Template link: https://one.one.one.one/dns/

Step Twenty Two – The last thing to do in this dialog is to click "Save".

Step Twenty Three – To verify that this change was effective, the DNS server assignment will no longer say "Automatic (DHCP)" and will now include the information that was entered.

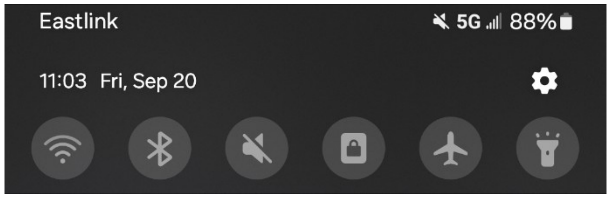


Wireless Public Networks and Virtual Private Networks

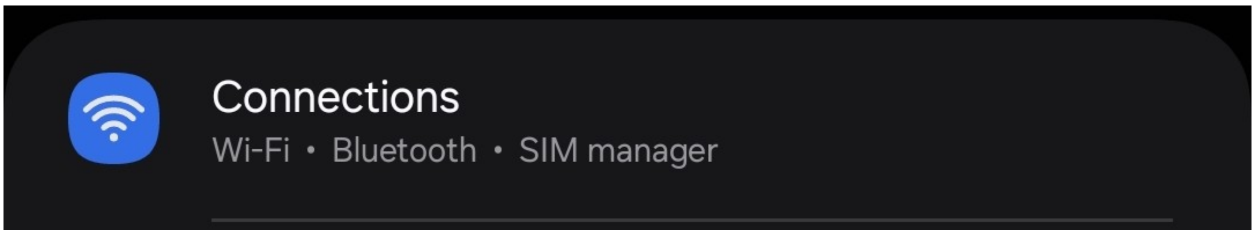
There are two solutions to avoid a privacy breach while connecting to the internet in public. One involves establishing a **Wireless Hotspot** on your mobile device, which involves creating a secure password and relying on a mobile device's network instead of connecting to a public one. The other option is to subscribe to a **virtual private network**, known as a VPN, which encrypts and masks the internet traffic, making it indiscernible to anybody that may be reading incoming activity from the computer.

Option 1: Protected Wireless Hotspot (Android)

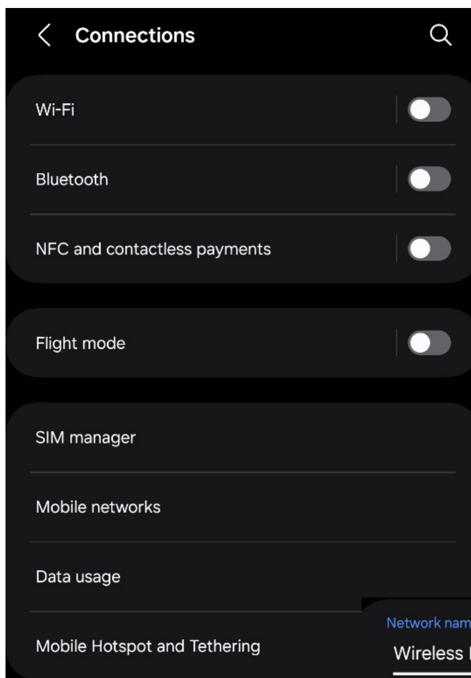
Step One – On a mobile device, open the **Settings** app. Drag down from the top of the screen to open the toolbar, and press the Settings button on the top right, indicated by a cog icon.



Step Two – Once the settings menu is open, tap "**Connections**"

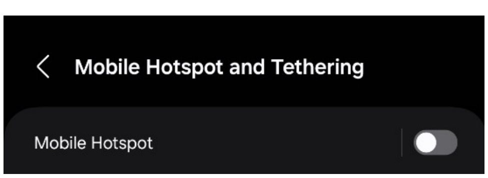


Step Three – Tap the "**Mobile Tethering and Hotspot**" section.

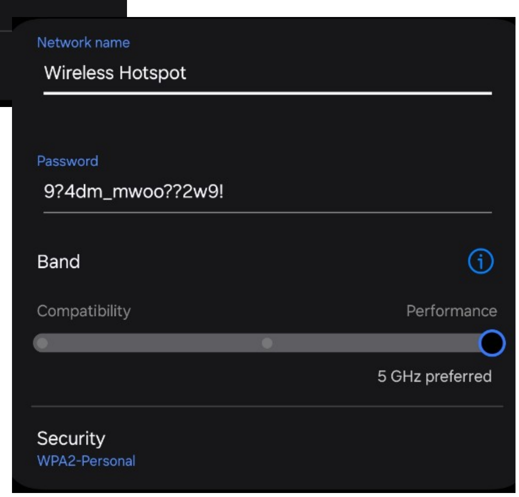


Step Four – Before turning on the hotspot, tap the section itself rather than the toggle. This will open the configuration settings for the network, where the user will set a name and password.

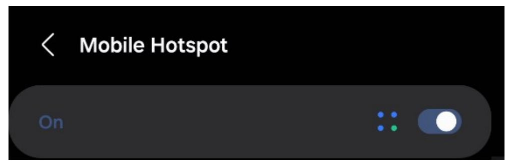
a.



b.

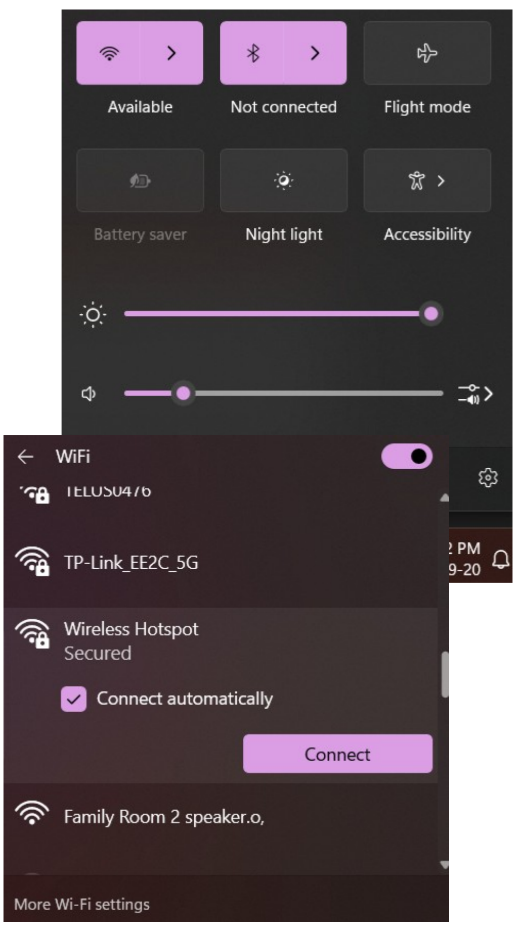


Step Five – Return to the previous page with the back arrow key on the bottom right of the screen. Now that the network has a password attached to it, it can be toggled on to activate the connection.

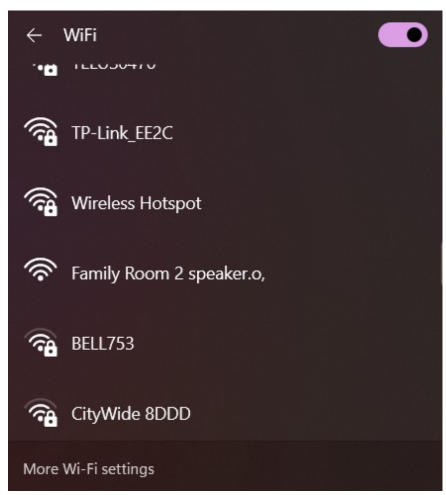


Step Six – On the laptop that's intended to connect, open the **Connections** tab by clicking on the internet, volume, and battery icons on the taskbar. Click on the arrow next to the Wi-Fi symbol and search nearby networks for the hotspot that was just created.

a.

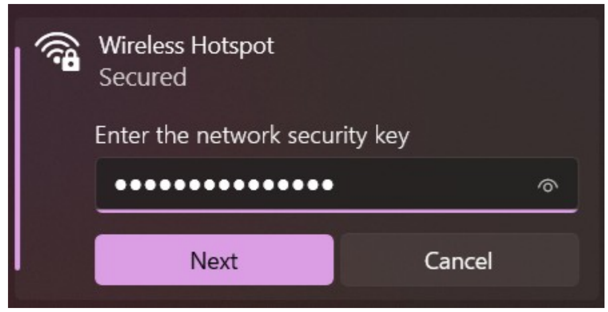


b.



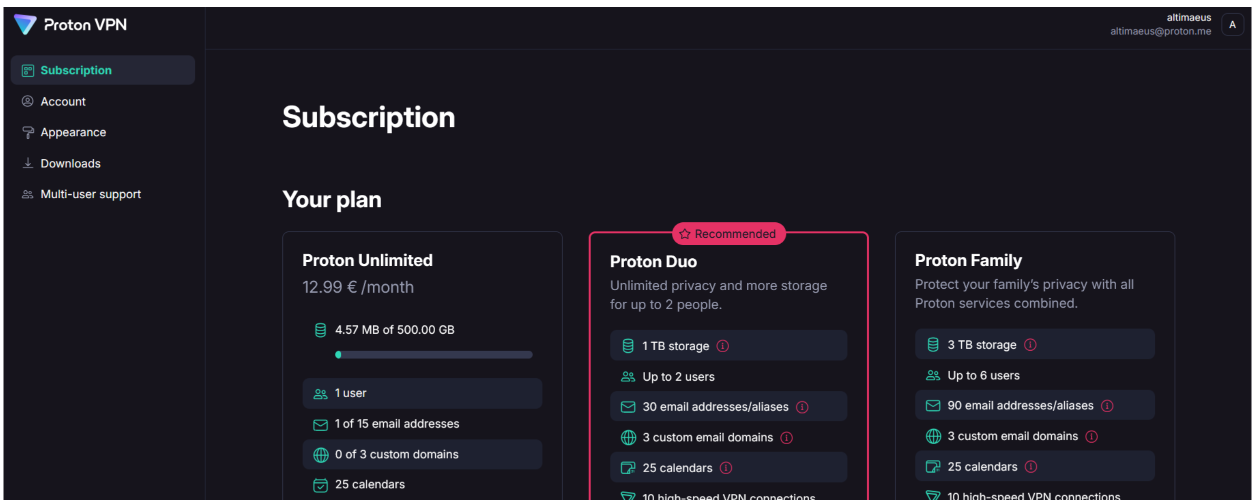
Step Seven – Once the correct network has been found and selected the network, press "**Connect**" and enter the new Wi-Fi password and click "**Next**". The computer should now be connected to the new network.

a. b.

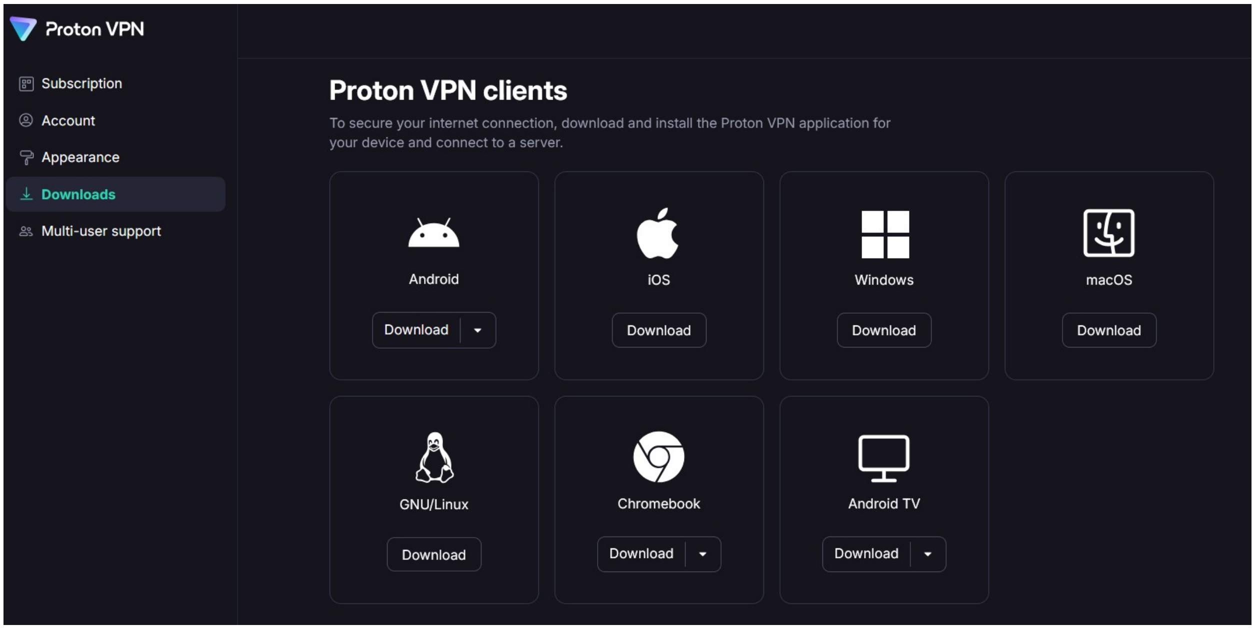


Option 2: Connect to a Public Network with a Virtual Private Network

Step One – Before going out to a location with a public network, have a virtual private network set up. The process to install one begins with visiting the website for your provider of choice. To download the VPN, click on the "Downloads" section where the intended operating system can be selected.

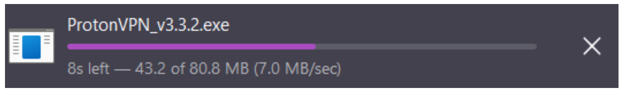


Step Two – In the Downloads section of the website, select the Windows application.



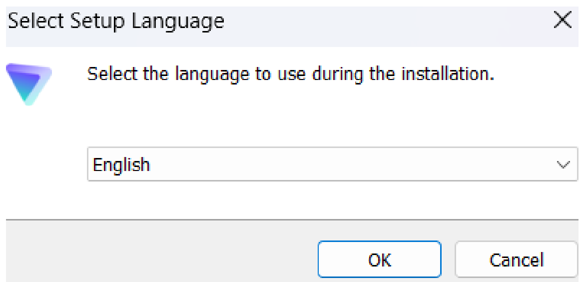
Step Three – Click "**Download Proton VPN**" in the new tab.

Step Four – Once the file is downloaded, click on it to begin the installation process.



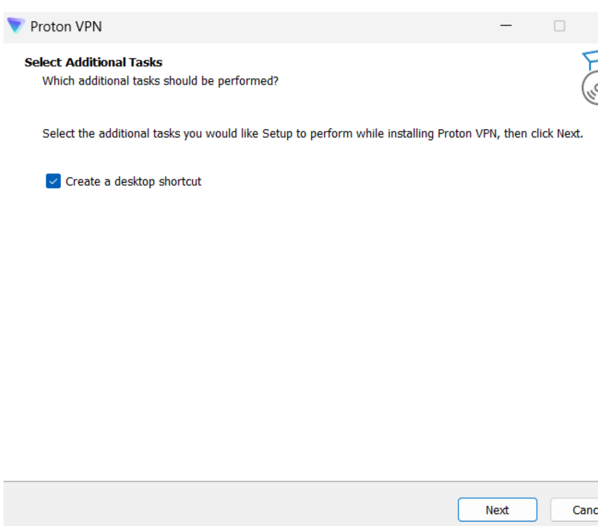
Step Five – The computer will ask for administrative permission to make changes to the device. This dialog cannot be shown in this article, as the permission overlay won't allow for anything above it. Select "**Yes**" to proceed.

Step Six – Select the required language.

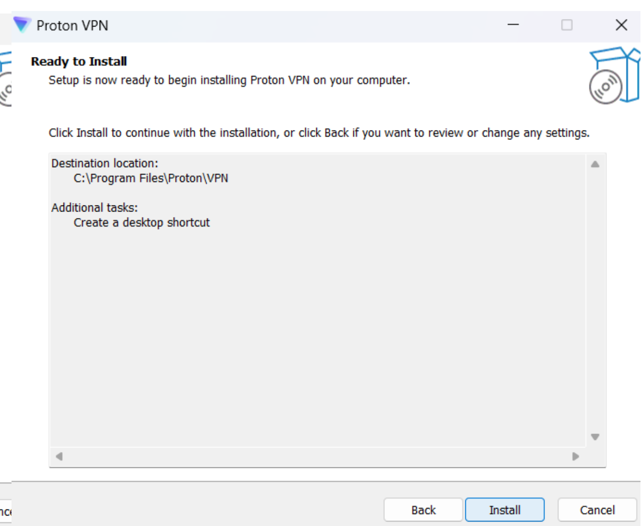


Step Seven – Select "**Next**" to create a desktop shortcut to the VPN service. Then, select "**Install**" to proceed.

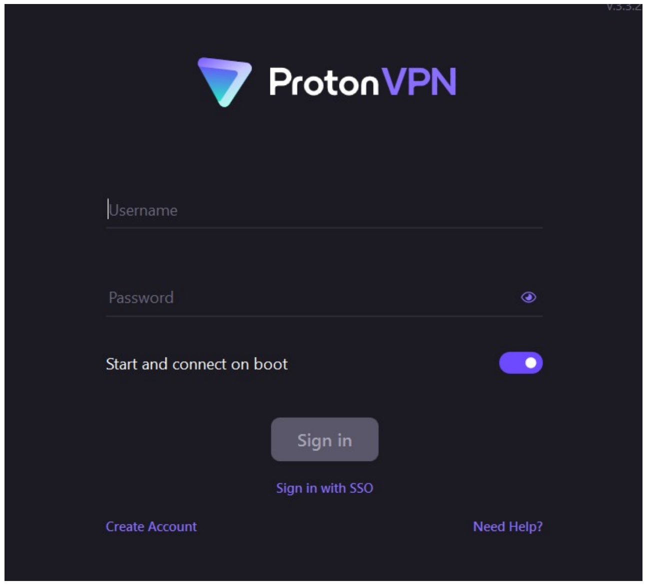
a.



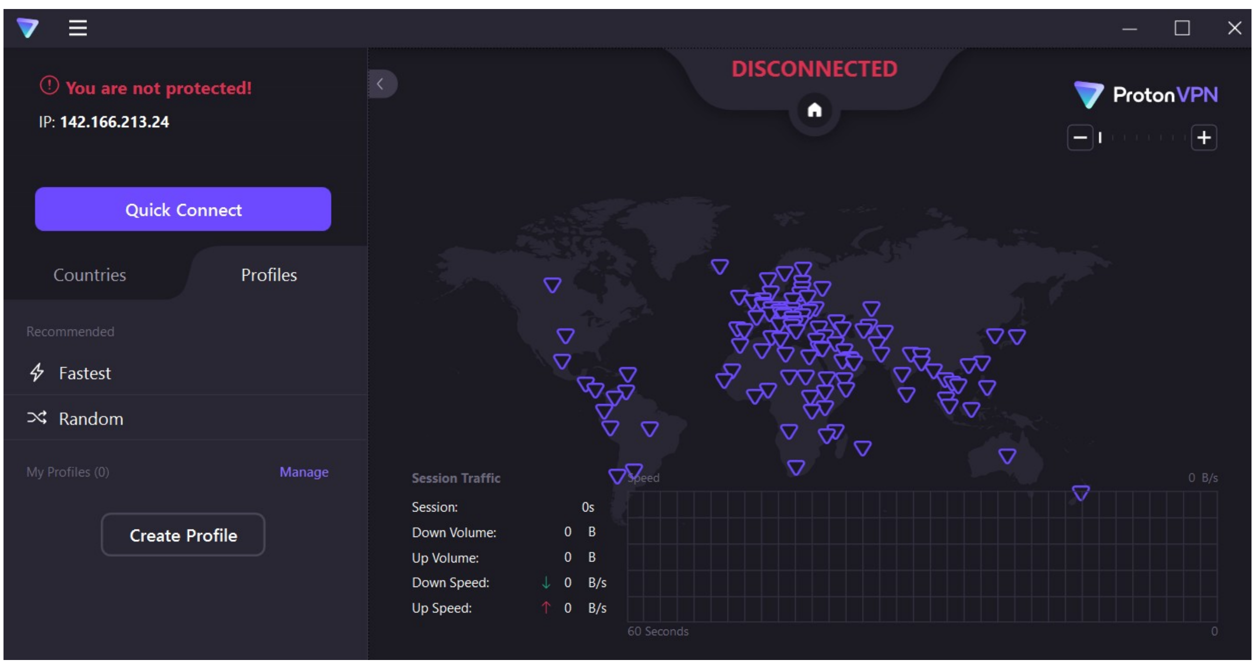
b.



Step Eight – Now that the installation is complete, open the VPN application and enter the required credentials.



Step Nine – Once the application has opened, click "Quick Connect" to find the fastest virtual network to connect to.



Use a Password Manager service

As time moves on, the nature of what is considered a strong password has evolved, and it has moved from a single word or set of numbers to a more complex set of words, letters, numbers, or special characters such as the exclamation point or brackets. If you are still using an account that can be accessed with an older form of password, or the same password that's been in place for several years, you will need to change your password. Due to the complexity of newer, stronger passwords, the set of characters can be difficult to remember, especially since a different password should be used for each account tied to your name or email address.

Out of convenience, many people choose to use a browser-based password manager that comes included with Microsoft Edge, Mozilla Firefox, or Google Chrome. These services generate passwords and automatically store them, which is more secure than other methods of password creation. However, they also store your passwords online through your account, allowing them to synchronize with another device when requested. Since they are features within a grander software, they may be also prone to vulnerabilities caused by interconnected services from your web browsing or app connections associated with your account. These passwords are also a target for physical theft, as even though they may be transmitted from device to device over AES-256 encryption, they're stored within the browser in a printable plaintext format and often don't require extra verification to view.

Standalone third-party software like Bitwarden, 1password, and Proton Pass offer safer encryption because of being end-to-end and are less prone to vulnerabilities caused by browser-related phishing or malware, alongside browser extensions that offer the same convenience without being maintained within a larger service model. The local storage is encrypted, and they offer a variety of backup options and secure emergency access tools ensure that the account can be recovered if access is lost, while also providing routine data breach scans to determine if a user has a compromised account.

If you have concerns with remembering the password of the account used to store this information, you can write it down so long as you treat the document as extremely confidential to yourself. Keep the document stored away from your device and secured in a file cabinet or other locked storage area. Keeping all your passwords written down and constantly needing to reference this document may lead to loss of the document and a potential breach of security, so it's best to stick to using a password manager service for all possible purposes.

Example 1: Bad vs. Okay vs. Good vs. Strong Password Creation

Bad Password – May2nd1987

a. Not many characters
b. A date that may be tied to your personal data.
c. No special characters.
d. As a result of having no special characters, it's likely that this password was created before enhanced security measures became normalized. If it's old enough, it's possible that it was involved in a data breach. (See **Phase III** for more information)

Okay Password – Computers1987!

a. Enough characters to be considered a strong password by detection software.

b. Contains a special character.
c. Still contains a personally connected date that is easy to guess.
d. Has a personal interest in the form of one word that is easy to guess.

Good Password – ajm_2mon5yr87pw-r?(sec)!

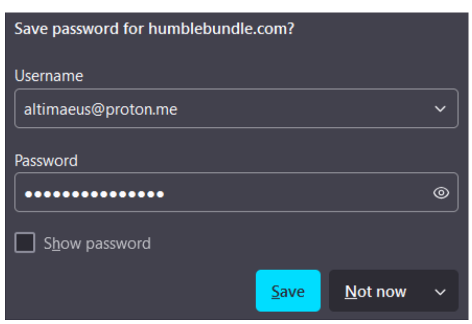
a. Identifiable characteristics that are only notable to the user for memorization purposes.
b. Long password with several special characters interspersed throughout. that makes it harder to crack by brute-force password cracking software, also known as credential stuffers.

Strong Password – r”092j”0r_#r2?0kr0)k4049fp!

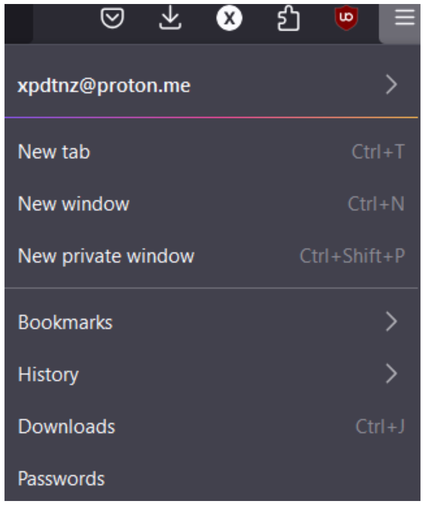
a. Long password with completely randomized characters. While not easy to memorize, this password is what you will see generated by a password manager.
b. Contains no identifiable personal information that can be used to potentially break into your account
c. Several special characters that make it harder for others to crack or steal.
d. You won’t be able to memorize hundreds of these kinds of passwords without extra-ordinary eidetic abilities.

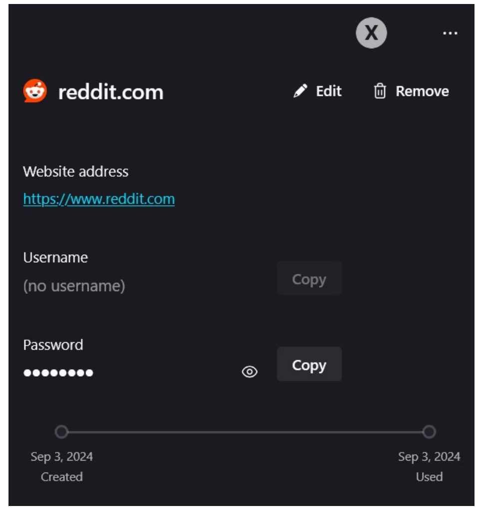
Example 2: Browser-Based Password Management (Firefox)

Step One – Register your email with the account you’re signing up for. A prompt will appear from your browser asking you to store your password. Press “**Save**” to store the password.

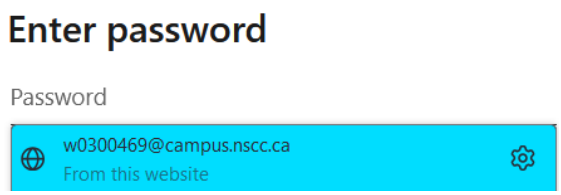


Step Two – Once stored, you can find them in the dropdown menu on the top right corner of the window. Find the account you wish to see the password for and click the eye icon to view it. To see the password on another device, click the icon associated with your browser account. In this example, it is the one marked with an X found next to your downloads and browser extensions. Then, you can click the “Sync Now” option. All your passwords will be downloaded to the new device and will be auto filled into the credential fields when you visit a website.

a. 

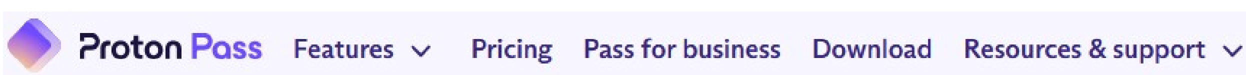
b. 

c. 

d. 

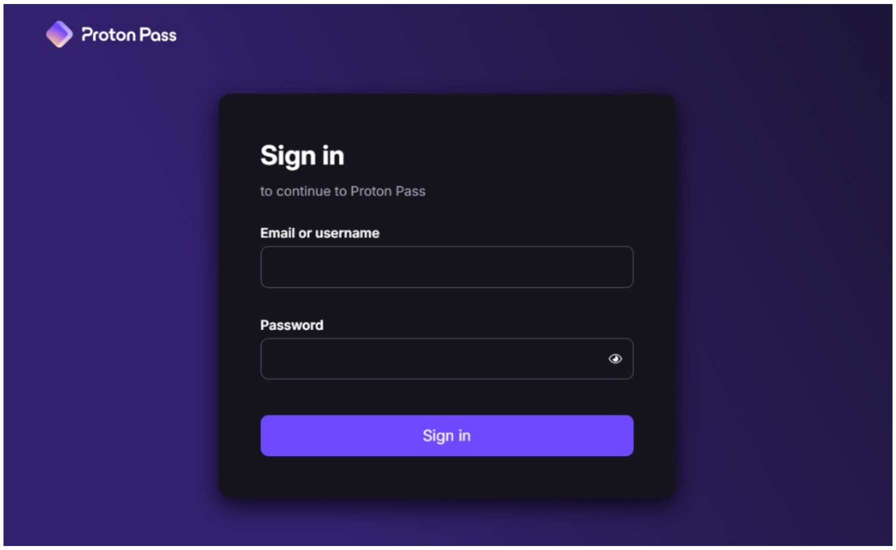
Example 3: Installing Proton Pass (dedicated password management software)

Step One – Visit the website for the password management software and select "**Download**". To download the desktop application, select your operating system. In this example, the option selected will be "**Download for Windows**".

a. 

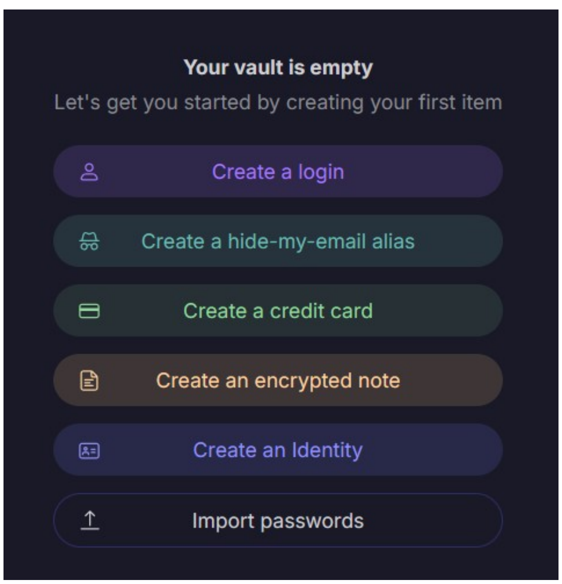
b. 

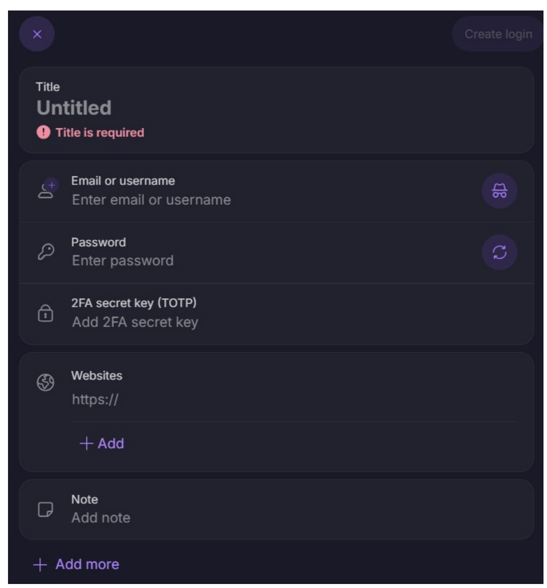
Step Three – The download dialog box will appear once the installation file has been downloaded. Click on it, and the installation process will automatically begin and you will be prompted to login.



Step Four –

Option 1 – Manual Entry: Once you log in, you will see an empty vault. This is where you can begin to add your passwords. Select "**Create a login**" and you will be able to enter your credentials. Once filled out, you can click "**Create login**" on the top right of the window. Repeat this for each password that you need stored.

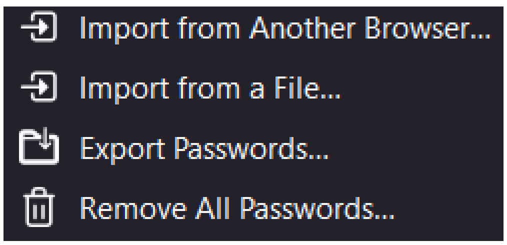
a. 

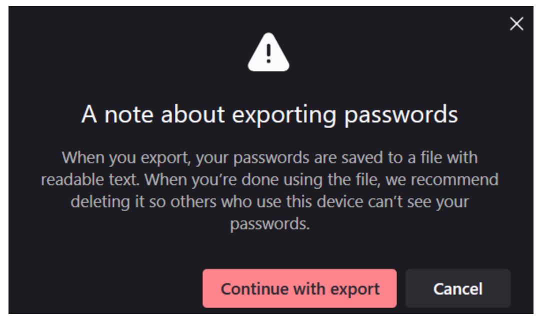
b. 

Option 2 – Import from Browser:

Step One – If you were already using a browser-based password manager, revisit the password page from **Example 2**. On the top right of the window, click the three dots (…) and select "**Export Passwords**". You will see a warning from **Firefox** regarding the readable text, where you will click "**Continue with Export**". Select a location to save the .csv file, preferably **not contained within OneDrive cloud storage** to avoid the risk of backups being created. Press "**Save**" to download the file to your requested location.

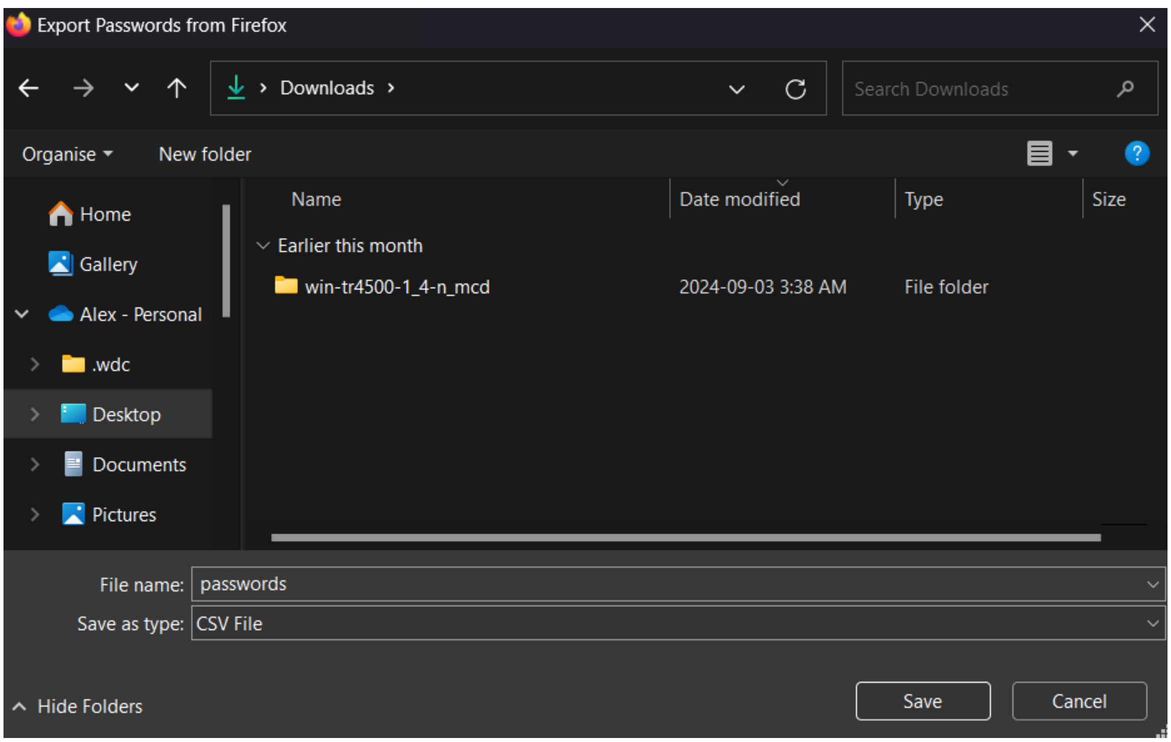
a.



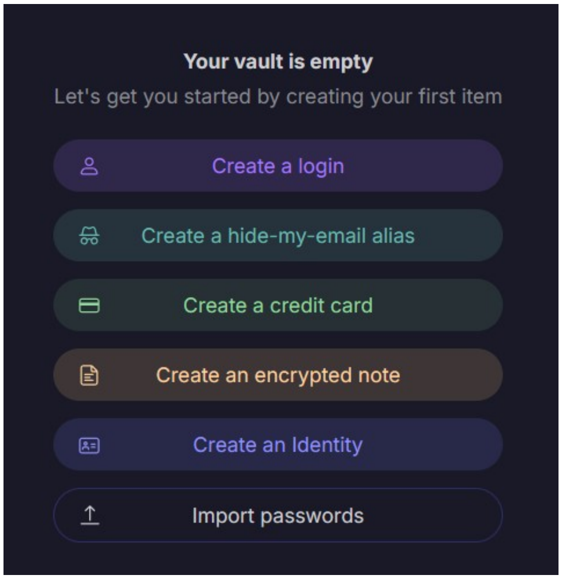


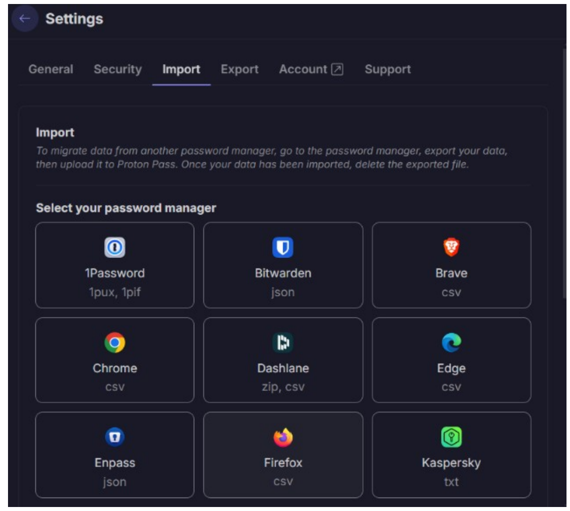
b.

c.



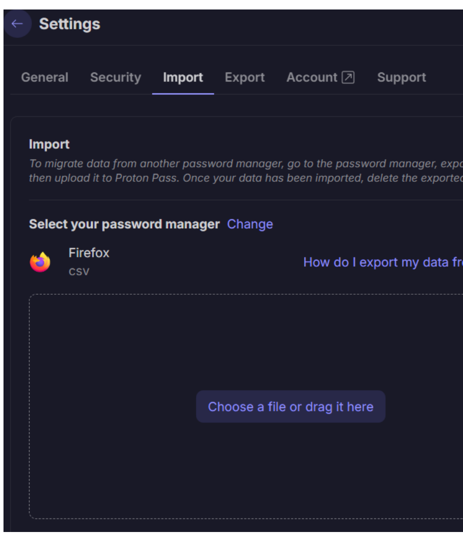
Step Two – Now that you have your passwords ready to import, return to your empty vault in the manager application. Select "**Import passwords**" and choose **Firefox**, since that was the browser used to export the passwords.

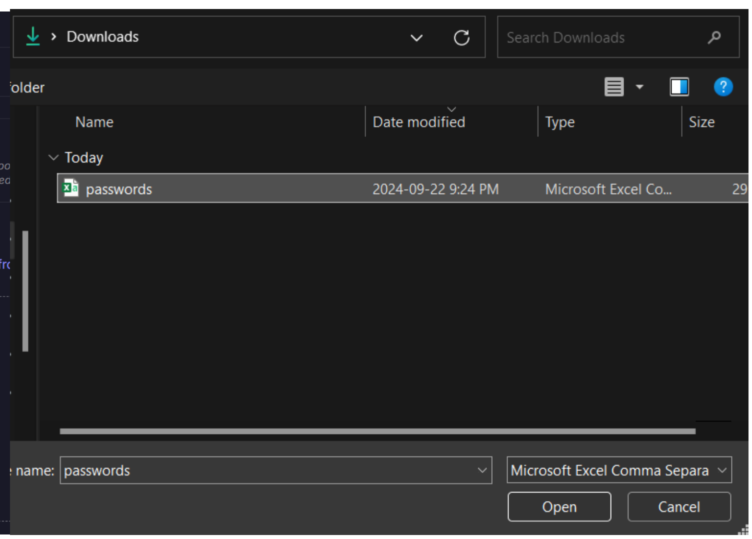
a. 

b. 

Step Three – To import the file, click "**Choose a file or drag it here**" and select the "**passwords.csv**" file that was downloaded. Select "**Open**" and now the passwords will appear in your vault.

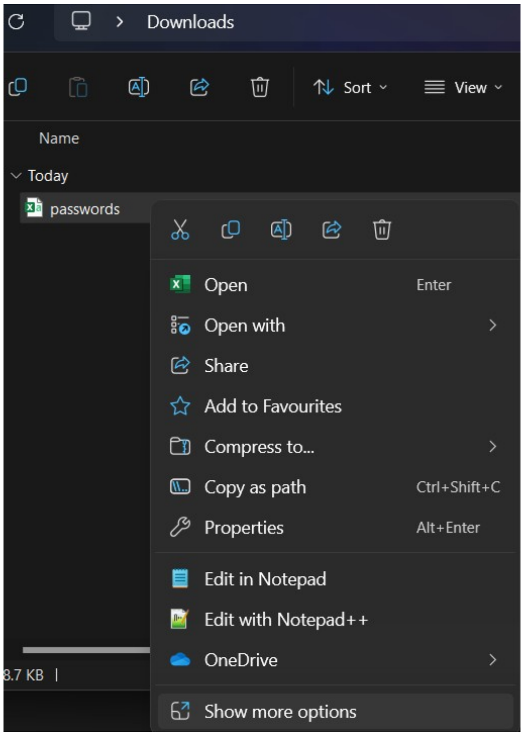
a. b.



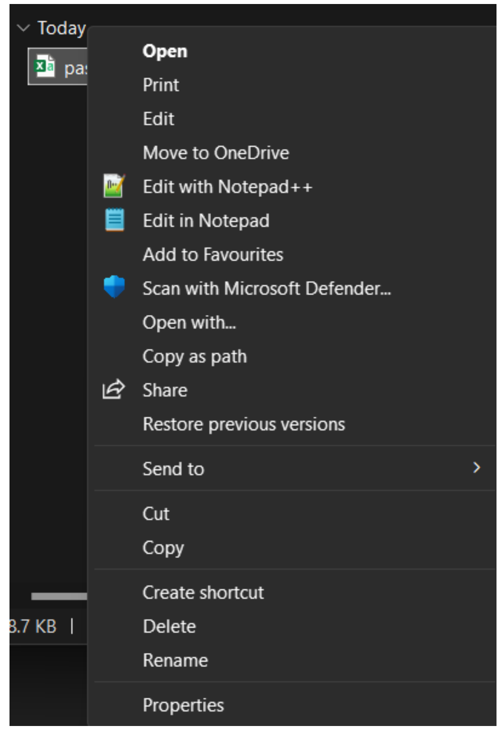


Step Four – Now that the files have been imported into a more secure location, delete the "**passwords.csv**' file from your computer. This can be done by opening the folder containing the file and using a right-click on the file. Press "**Show more options…**" and then click "**Delete**".

a.

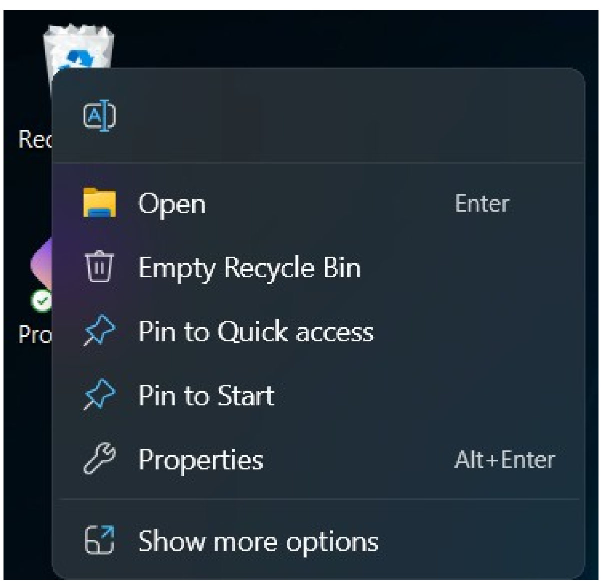


b.

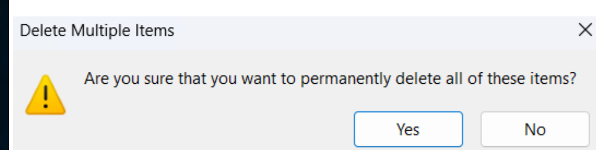


Step Five – Go to your desktop and empty the Recycle Bin. Right-click the bin and click "**Empty Recycle Bin**". It will ask you to confirm permanent deletion of the file. Just click "**Yes**" to complete the process.

a.

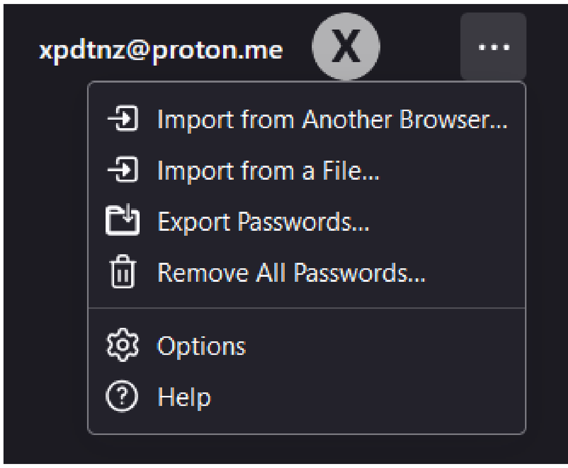


b.



Step Six – Return to **Firefox** and go back to the **Passwords** page. Click on the **three dots (…)** and click on "**Remove All Passwords…**". To complete the process, click on the confirmation checkbox that says “**Yes, remove passwords**” and then click “**Remove All…**”

a.



b.